

AI-POWERED WORKFORCE OPTIMIZATION

for Warehouse & Cargo Operations

Project Overview — Working Document (name not finalized)

1. The Problem

Warehouses and cargo operations deal with workloads that swing day to day and hour to hour. Some shifts are overwhelmed, others sit half empty. Most warehouses still schedule staff manually or with fixed rosters, which causes two costly mistakes:

- Overstaffing on quiet days — paying people to be idle.
- Understaffing on busy days — missed deadlines and rushed, expensive last-minute temp labor.

There's no affordable, AI-driven tool built for this problem in our region (Algeria/MENA). Existing solutions are enterprise-priced and not tailored to local warehouses or airport cargo operations.

2. The Idea

An AI tool that predicts how much work is coming, tells managers exactly how many workers are needed per shift, and builds the schedule automatically — optimized for cost and explained in plain language.

In one sentence: we predict the workload, then let AI build the cheapest schedule that still gets the job done.

3. How AI Is Used — Three Layers

Layer 1 — Forecasting

Predicts order/pick volume per shift and per day, based on historical patterns (day of week, seasonality, trends).
Answers: “how busy will we be next week?”

Layer 2 — Optimization

Takes the forecast and calculates the minimum-cost staffing plan that still covers demand — deciding how many regular vs. temporary/overtime workers per shift. This is real mathematical optimization (linear programming), not a guess or a lookup table.

Layer 3 — AI Assistant (LLM)

A chat assistant lets managers ask plain questions like “why do we need extra staff Thursday?” and get an answer grounded in the actual forecast/schedule data. It explains the system's own decisions rather than replacing them.

4. Who It's For

Primary target: warehouses, distribution centers, and airport cargo handling operations with variable daily workload and multiple shifts. Algeria-based operations are our first realistic customers — we can visit them, understand real constraints, and pilot with real data, which is far more valuable than any public dataset.

Airport cargo warehouses are a strong niche because their workload is driven by something predictable and external: flight schedules — a unique data signal competitors without an airport focus don't use.

5. Why Now / Why Us

- Labor is typically the single largest controllable cost in warehouse operations — even a modest scheduling improvement has visible ROI for a client.
- No existing local competitor is targeting this niche with an AI-first, affordable approach.
- We can start today with a working prototype using free/public and synthetic data, then refine it with a pilot client's real data.

6. Business Model (High-Level)

- Model: SaaS subscription, likely priced per warehouse/site or per number of employees scheduled through the system.
- Path to first customers: offer a free or low-cost pilot to 1–2 local warehouses in exchange for their historical data and feedback — used as a case study for future paying clients.
- Revenue scales with clients/sites added; cost scales mainly with hosting and AI usage, which is low at small scale.
- Longer-term expansion: other MENA countries, other logistics-heavy sectors (retail distribution, port operations, manufacturing).

We do not yet have real pricing, market size, or revenue figures. These should be defined only after real conversations with warehouse managers about their current labor costs and pain points — not estimated in advance.

7. Competitive Landscape (Brief)

- Logiwa, CognitOps, Takt, SAVIOM — enterprise-focused labor management platforms; expensive; not built for our market or for airport cargo.
- Instawork — on-demand staffing marketplace, not a forecasting/optimization tool.
- Our angle: affordable, local, cargo/airport-aware, and simple enough for non-technical managers to actually use day to day.

8. Technical Side (Explained Simply)

The prototype has four parts, each doing one clear job:

a) Data

No real warehouse data is freely available, so we generate realistic synthetic order-volume data (weekly patterns, growth trends, random demand spikes) to build and test the system. Swapping in a real client's actual order history later just needs the same simple format (date, shift, zone, order volume).

b) Forecasting Model

A machine learning model (trained with scikit-learn, a standard, free ML library) learns patterns in past order volume and predicts future volume per shift and day. It's validated against real held-out days to check accuracy before trusting its predictions.

c) Schedule Optimizer

A linear programming solver (a well-established mathematical optimization method) calculates the cheapest way to staff every shift — how many regular vs. temporary/overtime workers — while respecting labor rules (max hours, rest days).

d) Dashboard + AI Assistant

A simple web dashboard (built with Streamlit, a fast Python tool for data apps) shows the forecast, recommended schedule, and cost savings vs. a fixed/manual schedule. A chat feature powered by an AI assistant model lets managers ask questions in plain language, grounded in the actual data.

Current status: all four parts are built and working as a runnable prototype using synthetic data. Next step is testing it with a real warehouse's data.

9. What's Next — Team To-Do (In Order)

#	Action
1	Get the prototype running and understood by everyone on the team.
2	Prepare the pitch deck from this document (problem, idea, market gap, tech, demo, ask).
3	Identify 1–2 local warehouses or logistics companies to approach for a pilot conversation — even before building more features.
4	Based on pilot feedback, refine forecasting/optimization assumptions with real numbers instead of synthetic ones.
5	Only after that: define real pricing and business plan numbers.

Keep this document updated as the project evolves.